2023106120 Week 2 Assignment (2)

Question : 1)

2)

The Von Neumann architecture is a computer architecture in which instructions and data are stored in the same main memory and share the same communication pathway (bus) with the CPU.

Central Processing Unit

Control Unit

Input Device | Registers: MAR MDR PC ACC | Output Device

Memory Unit

Data program | Program Memory

First, the CPU fetches an instruction from the main memory using the Program Counter (PC). The Control Unit (CU) decodes the instruction to determine the required operation. If data is needed, it is fetched or needed, it is fetched from the same memory. The ALU performs the required materials, and the result is stored in a register or memory. The PC then moves to the next instruction, and the process continues until the program is completed.

b)
The Harvard architecture uses seprate memories and buses for instruction and data. Its main performance advantage is that CPU can fetch an instruction and access data at the same time because they travel through seprate buses.
It improves performance by using seperate memories and buses for instructions and data. This allows the CPU to fetch an instruction and access data at the same time, reducing delays, caused by sharing a single memory and a bus.

c)
Von Neumann architecture: General purpose computers such as desktop PCs and laptop.

Harvard architecture: Microcontrollers and digital signal processors used this architecture.

Question: 2)

a)
CISC is defined as a processor architecture that provides a large and a complex set of instructions, where individual instructions may perform several operations. Example: x86 processor family used by Intel and AMD.

RISC is defined as a processor architecture that uses a smaller and simpler set of instructions. Examples: ARM and RISC-V processor families.

b)

CISC	RISC
i) It is large and complex	i) It is smaller and simpler.
ii) It instructions may perform multiple operations.	ii) It instructions generally perform multiple operations.
iii) It Instruction length often variable.	iii) It instruction usually fixed or more uniform.
iv) It has more complex control logic.	iv) It has simpler control logic.

c)

CISC: Example → x86 processor family - Intel Core and AMD Ryzen processors.

Advantage: It can perform complex operations using fewer instructions, which can result in compact program code and good compatibility with existing software.

Disadvantage: The complex instruction set requires more complicated hardware and control logic, which can make processor design and instruction execution more complex.

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RISC: ARM Processor Family

Advantages: It's simpler and regular instructions make pipelining and efficient processor design easier, often providing good performance with relatively low power consumption.

Disadvantages: A complex task may require more individual instructions, potentially increasing program size and placing greater demands on compiler.